

Week 2 - Assignment 2

ML and GenAI with Python

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Q1) A bank wants to predict whether a loan application will be approved or rejected. Is this a classification or regression problem? Which algorithm among Decision Tree, KNN, and Linear Regression would be most suitable?

Ans 1) A bank wants to predict whether a loan application will be approved or rejected. This is a classification problem because the output belongs to a fixed set of categories: Approved or Rejected. The model is not predicting a numerical value but assigning the application to one of the predefined classes.

Decision Tree would be the most suitable algorithm because it can handle both numerical and categorical data and provides clear decision rules. For example, the tree can make decisions based on factors such as income, credit score, employment status, and loan amount. It is also easy to understand and interpret compared to other algorithms.

Q2) Explain the difference between a feature and a label using a real-world example.

Ans 2) In machine learning, a feature is an input variable used by the model to make predictions, while a label is the output or target value that the model tries to predict. eg) Suppose we want to predict the price of a house.

Features:

- Area of the house
- Number of bedrooms
- Location
- Age of the house

Label:

- House price

The model learns the relationship between the features and the label and then predicts the price of new houses.

Q3) A Decision Tree predicts that a customer will purchase a product. Explain how the model reaches this prediction.

Ans 3) A Decision Tree makes predictions by splitting data into branches based on conditions. Each answer moves the customer down a different branch of the tree. Eventually, the customer reaches a leaf node that contains the final prediction. Thus, the model reaches its decision through a sequence of simple rules learned from historical data.

Q4) Why is KNN considered a supervised learning algorithm? Can KNN perform classification if labels are not available?

Ans 4) KNN (K-Nearest Neighbors) is called a supervised learning algorithm because it learns from training data that already contains known labels or outcomes. When a new data point is given, KNN finds the K nearest labeled data points and predicts the output based on their labels.

KNN cannot perform classification if labels are not available . KNN requires labeled data to perform classification. Without labels, the algorithm cannot determine which class a new data point belongs to because it has no reference for comparison. Therefore, labeled training data is essential for KNN classification.

Q5) Suppose a KNN model uses: $K = 5$ and the nearest neighbors are: Yes, Yes, No, Yes, No What will be the prediction and why?

We have- $K = 5$, Nearest neighbors: Yes, Yes, No, Yes, No

Prediction - The prediction will be **Yes**.

Explanation - KNN uses majority voting to determine the final class.

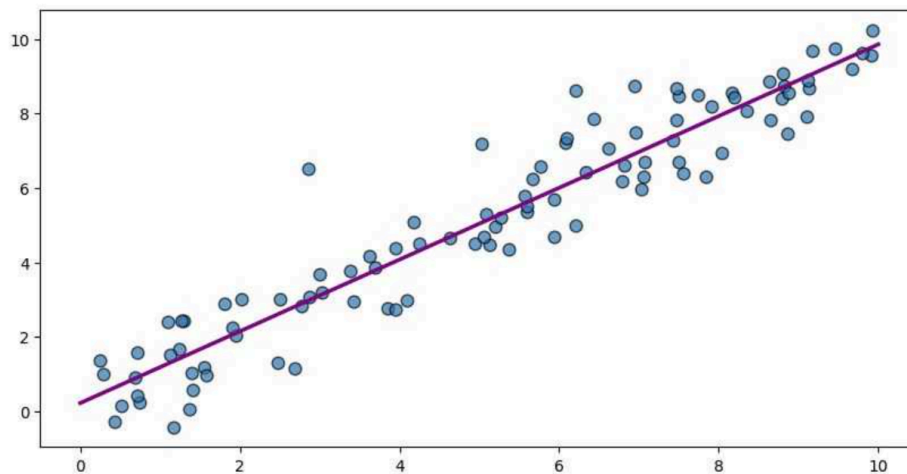
Count of neighbors:

- Yes = 3
- No = 2

Since the majority of the nearest neighbors belong to the "Yes" class, the model predicts **Yes** for the new data point.

Q6) Explain the equation of Linear Regression. What do x, y, m, and b represent?

Ans 6) The equation of Linear Regression is: $y=mx+b$



x = Independent variable or input feature.

y = Dependent variable or predicted output.

m = Slope of the line, which indicates how much y changes when x changes.

b = Intercept, which represents the value of y when x equals zero.

If x represents the number of hours studied and y represents exam marks, the equation can be used to predict marks based on study hours. Linear

Regression finds the best-fitting straight line that represents the relationship between the input and output variables.

Q7) What is correlation? $y = mx + b$ What does each of the following values indicate? +1, 0, -1

Ans 7) Correlation is a statistical measure that describes the strength and direction of the relationship between two variables.

Its value ranges from -1 to +1.

+1 (Perfect Positive Correlation)- A value of +1 means both variables move in the same direction.

Example: As study hours increase, exam marks also increase.

0 (No Correlation) - A value of 0 means there is no relationship between the variables.

Example: Shoe size and IQ.

-1 (Perfect Negative Correlation) - A value of -1 means the variables move in opposite directions.

Example: As speed increases, travel time decreases.

Correlation helps data scientists understand how strongly variables are related before building predictive models.

Q8) A disease detection model produced the following result:

Actual Condition: Disease Present, Prediction : Healthy. What type of prediction is this? Why can this be dangerous?

Ans 8) This is a False Negative (FN) prediction. A False Negative occurs when the model predicts that a person is healthy even though the disease is actually present. This type of error can be very dangerous because:

- The patient may not receive medical treatment.
- The disease may continue to spread or worsen.
- Doctors may assume the patient is healthy.
- Serious health complications can occur due to delayed diagnosis.

In healthcare applications, minimizing False Negatives is often very important because missing a disease can have severe consequences.

Q9) What is the purpose of a Confusion Matrix? How is it different from a Correlation Matrix?

Ans 9) A Confusion Matrix is a table used to evaluate the performance of a classification model. It compares actual values with predicted values and helps identify different types of predictions:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Using these values, metrics such as accuracy, precision, recall, and F1-score can be calculated.

Confusion Matrix	Correlation Matrix
Used to evaluate classification models	Used to measure relationships between variables
Compares actual and predicted outcomes	Compares variables with each other
Contains TP, TN, FP, FN values	Contains correlation values from -1 to +1

Used in machine learning evaluation	Used in statistical analysis
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Q10) Compare the following algorithms in one or two lines each: Decision Tree, KNN, SVM, Linear Regression, Neural Network. Focus on the main idea behind each algorithm.

Ans 10) Decision Tree - A Decision Tree creates a tree-like structure of questions and decisions. It predicts outcomes by following branches based on feature values and is easy to understand and interpret.

K-Nearest Neighbors (KNN) - KNN classifies a new data point by examining the nearest labeled data points and using majority voting or averaging. It is simple and effective for small datasets.

Support Vector Machine (SVM) - SVM finds the optimal boundary that separates different classes with the maximum possible margin. It performs well on high-dimensional datasets and complex classification problems.

Linear Regression - Linear Regression predicts continuous numerical values by finding the best-fitting straight line between input and output variables. It is commonly used for forecasting and trend analysis.

Neural Network - A Neural Network consists of interconnected layers of artificial neurons that learn patterns from data. It is capable of solving complex problems such as image recognition, speech processing, and natural language understanding.